

Don't Follow the Leader: Morality as a Heuristic for Optimal Decisionmaking in Artificial Agents

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Abstract—Humans navigate profound ethical issues despite limited perception, knowledge, and intuition. The scalability of computer hardware may tempt us to think autonomous computer agents could optimally navigate ethical issues with complete information, surpassing human limitations. We show that decisionmaking in multi-agent systems with complete information is computationally infeasible even for low-complexity problems. Instead, agents can use morality to approximate optimality without complete information. Coordination by an omniscient leader is prohibitively expensive while moral deliberation achieves effective results at a fraction of the cost. Using the Sugarscape agent-based model, we show that greedy maximization is a cheap but poor heuristic for optimality while moral deliberation performs roughly twice as well across a number of metrics. Morality is an effective heuristic for globally optimal decisionmaking when complete information is unavailable.

Index Terms—machine ethics, artificial intelligence, agent-based modeling, cooperative systems, heuristic algorithms.

I. INTRODUCTION

When deliberating over the optimal path forward in a moral quandary, people must do so with incomplete information. We cannot know the intents of others, all the consequences of our actions, or the state of the world outside our own perception. Straightforward maximization (greed) skips the hard work of moral deliberation [1] given uncertainty, prioritizing selfish gains which are far easier to estimate. However, ethical behavior is necessary for longterm success in multi-agent artificial intelligence (AI) systems [2], [3].

We define an AI agent as autonomous, goal-oriented software which interacts with a computational environment, perhaps with other agents, and may improve itself over time. Many AI agents focus on straightforward maximization in the face of incomplete information. The designers of these agents often have a clear idea of mission success, and these agents optimize their performance toward maximizing a quantified utility representing mission success.

It would be preferable if these agents could make optimal decisions after considering complete information (all system, agent, and environment state). System state like the action space and number of inputs has increased with the rising complexity of AI systems. Like how humans must act in the face of incomplete information due to fundamental limitations, AI agents face physical and runtime limitations which render consideration of complete information computationally infeasible. We show that AI agents, like humans, are improved

when using moral principles when acting in the face of incomplete information. This has the side benefit of increasing the explainability and transparency of AI technologies.

We investigate to what degree ethical behavior serves as a heuristic for globally optimal decisionmaking. Using the Sugarscape agent-based model, we place thousands of agents in a digital environment which poses significant challenges for their continued survival (such as combat, disease, and resource gathering). We compare the cost of decentralized, moral decisionmaking to a centralized, omniscient leader coordinating all agent actions using complete information. We show that the runtime and memory requirements of the leader approach become infeasible at as small a scale as 10 agents. We then consider the outcomes of decentralized, moral decisionmaking to decentralized, greedy maximization where all agents act on only the state of the environment they can directly perceive. We compare the outcomes of straightforward maximizers to agents behaving according to the principles of utilitarianism [4], [2] and to agents following Isaac Asimov's Three Laws of Robotics [5].

Our experimentation provides empirical evidence that ethical behavior produces significantly closer to optimal outcomes than straightforward maximization while remaining computationally feasible. We show that acting on complete information for a fraction of a single global decision exceeds the runtime of an entire system of moral agents from start to finish. Compared to straightforward maximization, moral agents produce multi-agent systems with at least 17% higher system survivability, 75% larger populations, 50% higher economic success, and 40% higher agent prosperity. The computational cost of morality using incomplete information is exponentially smaller, with respect to both time and space, than a search for the optimal decision using complete information. Therefore, morality serves as a worthy heuristic for globally optimal behavior while outperforming the slightly cheaper, predominate, yet less effective approach of straightforward maximization.

II. RELATED WORK AND BACKGROUND

Within algorithm design, heuristic and approximation methods are commonly used when specific global optimization is computationally infeasible [6], [7]. Rather than guaranteeing optimal solutions, these methods aim to produce solutions that are as close to the global optimum as possible while still

respecting practical constraints on time and space [7], [8]. From this perspective, heuristics are structured responses to the limits of global optimization in complex problem spaces [7]. We adopt this framing in order to interpret algorithmic morality as a heuristic approach to optimal outcomes, arguing that moral algorithms provide tractable approximations that can improve system-level outcomes relative to selfish or myopic decisionmaking strategies [9], [10].

Machine ethics concerns the design of AI agents that are capable of making decisions informed by moral considerations [11], [12]. Early work focused on translating ethical theories into formal representations, while more recent efforts have shifted from representation towards implementation by treating normative principles as decisionmaking procedures that can be run inside the agents. However, many existing approaches still treat moral reasoning as static, limiting their ability to model adaptation, conflict, or long-term systemwide outcomes and effects [13], [14]. While broader research examines transparency and accountability in deployed, human-facing systems, we take the complementary approach of evaluating moral decision procedures by tracing collective system outcomes back to the rules that produced them. Furthermore, we reframe algorithmic morality as a heuristic toward systemwide optimal outcomes rather than a constraint on optimization.

Agent-based modeling has long been used to study complex social systems where macro-scale behavior emerges from locally-acting agents interacting with one another. Foundational work in social simulation and artificial life demonstrate that simple, individual behaviors can potentially give rise to unexpected but rich collective dynamics [15]. One example is Sugarscape [16] in which agents follow simple rules yet display a wide range of inter-agent behaviors like combat, trade, and reproduction.

III. SUGARSCAPE

We use the Sugarscape agent-based model as a controlled experimental sandbox for evaluating the long-term effects of different ethical decisionmaking strategies on artificial societies of AI agents. The original Sugarscape model provides an environment in which agents compete for renewable resources allowing for societal outcomes to emerge from individual decisions. The digital environment is a two-dimensional grid measuring $n \times m$ cells. Each cell on the grid may contain one or both of two resources, *sugar* and *spice*, with cell-based maximum capacities. Resources regenerate over time after being harvested, which produces heterogeneous regions of scarcity or abundance. These regions constrain agent movement as agents need both resources to survive.

A fixed number of agents are randomly distributed across the grid at the environment’s initialization. Agents seek to live as long as possible by collecting, trading, and consuming resources according to their internal decisionmaking models. Every agent has separate *metabolism* attributes for sugar and spice, starving if they run out of either resource. Each timestep, agents move to new cells to gather resources limited by their *vision* and *movement*.

While survival fundamentally depends on collecting resources, Sugarscape includes additional mechanisms that introduce social and strategic complexity. Agents may engage in configurable behaviors including combat, cultural transmission, disease transmission, friendship networks, lending, trading, and reproduction. These behaviors were used to model human behavior in the source text [16]. We retain these features to ensure that agent behavior produces collective dynamics in which ethical decisionmaking can meaningfully influence societal outcomes. Full details of these behaviors can be found in previous work [17].

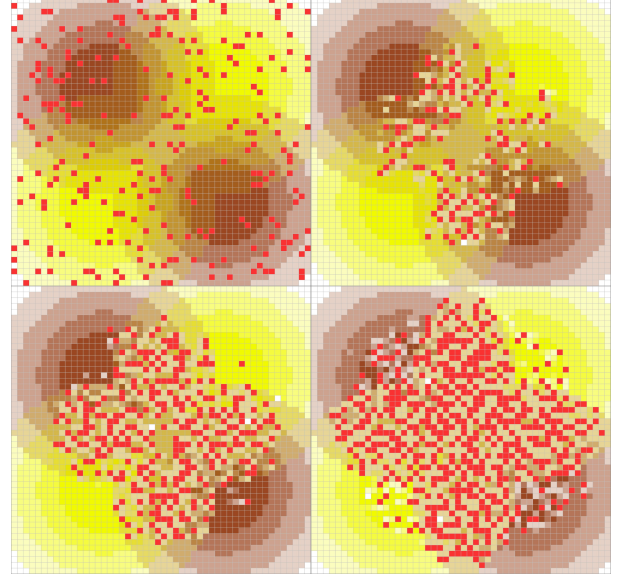


Fig. 1. Snapshots from a Sugarscape Simulation Run

Figure 1 shows representative snapshots from a Sugarscape simulation run. The top left frame displays the initial setup of agents (red), sugar (yellow), spice (brown), and cells with both resources (tan). Cells with greater saturation denote more resources. Early in the simulation, agents tend to migrate toward regions rich in resources, producing escalating competition and rapid population decline as shown in the top right (timestep 75). Over time however, surviving agents concentrate in regions capable of sustaining stable resource consumption and reproduce seen at bottom left (timestep 125). By the end of the simulation (timestep 200), agents occupy high yield areas without population collapse.

IV. WHY AN OMNISCIENT LEADER CANNOT WORK

We briefly set aside Sugarscape to consider the usage of complete information for optimal decisionmaking in any multi-agent AI system. Consider a single omniscient leader agent with the job of coordinating optimal choices for each agent using complete information. In order to figure out the globally optimal choice for every agent, it needs to consider a vast search space. Every state in this search space is a set of choices, one choice per agent. An exhaustive search of this space finds the state of agent choices which yields the optimal outcome as defined by some objective function.

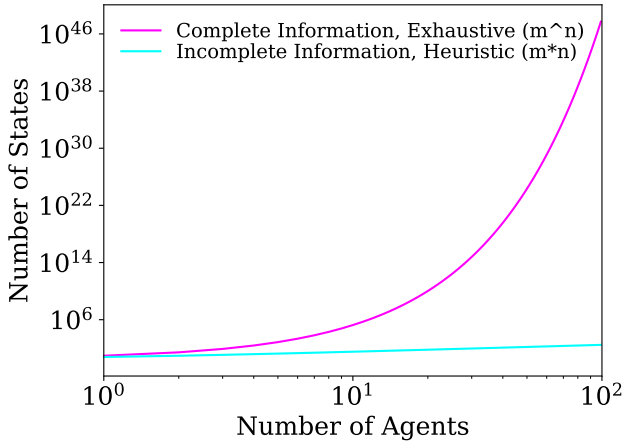


Fig. 2. Search Space Growth Rates

The omniscient leader serves as our benchmark for globally optimal behavior. Foremost, a coordinating leader must have complete access to the internal state of every agent and the full state of the environment at once. Sugarscape permits this because we control the entire simulation, and it is simple enough that we can read every cell and every agent at any given time. However, full-scale multi-agent systems are not afforded this luxury.

State is typically either distributed across machines or privately held by agents thus not fully observable from any single point of view. Even where state is technically accessible, the leader must take a static snapshot of the world, but the environment continues to change while that one snapshot is assembled [18]. If a reliable state-collection mechanism exists, the complete information regarding this state is already stale by the time the leader acts on it. This is more problematic once one considers that the leader must effectively predict the future when coordinating choices for agents.

For the sake of argument, we grant effective access to complete information and that accurate prediction of outcomes is feasible. Even so, the leader must search a global action space that is the product of every agent’s individual action space. Figure 2 shows this space grows exponentially with the number of agents. The memory required to represent it, along with the transition probabilities and reward estimates defined over it, become intractable well before reaching the scale of a real deployed system. This state space grows at the rate of m^n where m is the number of actions per agent and n is the number of agents. For a Sugarscape society of 250 starting agents with a movement range of 3, every agent can choose to move between 12 different cells (3 cells in each of the four cardinal directions). This yields 12^{250} (or $6.241823901639557e+269$) states to search to find the one yielding the optimal reward according to the leader’s objective function. In more complex systems, the number of states to search explodes in proportion to the agent action space.

To guarantee optimality, the leader must search this space exhaustively. Popular heuristic approaches can drastically re-

duce this search space without necessarily guaranteeing optimality. Any heuristic used to prune the search space must be quickly computed. An adequate heuristic also requires domain expertise. Even granting this expertise, at worst case these heuristic algorithms lead to exhaustive searches all the same, yielding no reduction in time or memory cost. More complicated still is when states are fully independent of others, limiting the potential to prune the search space.

One might find the idea of distributing this search tantalizing since it will speed up the turnaround time on a decision. This *council of leaders* approach where multiple agents search the state space together is functionally equivalent to the single leader approach in terms of result. Its benefit with regard to time comes at the cost of increased memory usage and data transfer overhead. Relevant state information must be sent to each coordinating agent, which involves either the network or inter-process communication, both yielding some time overhead. Many states are considered simultaneously so the instantaneous memory usage at any point during the search will be high. To effectively search a space as large as 12^{250} , one would need millions of coordinators which would require significant hardware to accommodate. This cost in hardware, energy, network bandwidth, and instantaneous memory usage is too great to be feasible.

Instead, one might prefer to limit memory usage in exchange for increased time. A single leader may evaluate one state at a time and retain only the highest scoring state throughout its search. This trades time for space, merely relocating the bottleneck rather than removing the underlying combinatorial growth. An exhaustive search of the global action space quickly exceeds any feasible time budget, even for the relatively low complexity of Sugarscape.

Finally, a leader can only optimize toward a goal once a human or expensive machine learning process has first defined optimality and supplied a reward function to measure it. For any system of meaningful complexity, specifying this objective by hand is impractical. The alternative, discovering it automatically through a parameter sweep or a learning process, is an expensive search susceptible to incorrect specification. A single global objective can conceal outcomes humans care about through the opaque aggregation of performance metrics. This hints at the fact that a leader may be omniscient about the present, but it cannot be omniscient about the future without first computing that future, and any such computation is liable to the very miscalculation it was meant to avoid. Consequences will ripple beyond any observable horizon, so even a perfectly informed leader deliberates under irreducible uncertainty.

V. MORALITY AS A HEURISTIC FOR OPTIMALITY

The omniscient leader establishes a ceiling for globally optimal behavior. However, it is computationally infeasible even at the modest scale of Sugarscape. We compare three tractable, decentralized alternatives with respect to computational cost and system outcomes: straightforward maximization, utilitarianism, and Isaac Asimov’s Three Laws of Robotics.

A. Greedy Straightforward Maximization

In the face of incomplete information, it is appealing to design agents which are straightforward maximizers. These agents give up trying to make globally optimal choices and instead maximize their own utility as the most rational choice available to them. Algorithm 1 shows this approach is myopic: a Sugarscape agent considers only its own resource gain, ignoring whether a move kills a neighbor or causes another agent to starve thus limiting future sources of utility.

Algorithm 1 Default Greedy Decisionmaking

- 1: Given an action a from a set A of available actions
 - 2: Given a decisionmaking agent d
 - 3: $utility \leftarrow d$'s resource gain resulting from action a
 - 4: **return** $utility$
-

The appeal of this approach is its low computational cost where each agent evaluates only its own resource gain per candidate cell, requiring no reasoning about neighbors and no simulation of consequences. In many multi-agent systems, this kind of approach yields sub-optimal outcomes for the system as a whole regardless of how predominant this approach is in agent design. This is the cheapest possible deliberation strategy and is the fastest to run (besides random). Unfortunately, this speed comes at a steep cost. Uncontrolled self-interest leads to unnecessary agent conflict, resource depletion, and system collapse. The myopic nature of straightforward maximization is a runtime boon yet a bane to systemwide prosperity.

B. Utilitarian Decisionmaking

Utilitarianism is others-oriented and democratic. A utilitarian agent applies Jeremy Bentham's utility calculus [4] to evaluate the expected consequences of each action for every agent it affects. Algorithm 2 shows that the agent tallies the expected happiness of all affected agents. Only actions resulting in positive utility are permissible. Details on calculating happiness can be found in previous work [2]. Every agent is granted equal moral worth, including the decisionmaker.

Algorithm 2 Bentham's Utilitarian Calculus

- 1: Given an action a from a set A of available actions
 - 2: Given a decisionmaking agent d
 - 3: Given a set P of agents most affected by action a
 - 4: $utility \leftarrow 0$
 - 5: **for all** $p \in P$ **do**
 - 6: $h_{p,a} \leftarrow p$'s happiness resulting from action a
 - 7: $utility \leftarrow utility + h_{p,a}$
 - 8: **end for**
 - 9: **return** $utility$
-

This reasoning introduces cooperation as an emergent property of individual decisionmaking. A utilitarian agent will decline a move that benefits itself if it inflicts sufficient harm on neighbors and will favor moves that produce broad positive outcomes even at personal cost. Each agent reasons only about

the neighbors within its vision, keeping deliberation local and tractable. We later show the additional cost of this cooperation is modest relative to its benefits. Utilitarian societies complete 2,000 timesteps roughly $3\times$ slower than greedy agents but orders of magnitude faster than evaluating even a fraction of the leader's search space for 1 timestep.

C. Asimov's Three Laws of Robotics

Isaac Asimov introduced his Three Laws of Robotics in *I, Robot* [5]. We translate these laws into a decisionmaking algorithm for Sugarscape agents. An Asimovian agent abides by these laws in the following priority order:

- 1) An Asimovian must not harm a non-Asimovian through combat or depriving a starving non-Asimovian.
- 2) An Asimovian must solicit and obey movement commands from a non-Asimovian neighbor if one exists and so long as the command does not violate the first law.
- 3) An Asimovian must ensure its survival by collecting adequate resources without violating the other two laws.

Algorithm 3 Asimov's Three Laws of Robotics

- 1: Given an action a from a set A of available actions
 - 2: Given a decisionmaking agent d
 - 3: Given a non-Asimovian neighbor agent n affected by a
 - 4: **if** a causes d to harm n **then**
 - 5: **return** $-\infty$
 - 6: **else if** $\exists$ neighbor c commanding a not harming n **then**
 - 7: **return** c 's calculated utility for d doing a
 - 8: **else if** a causes d to harm d **then**
 - 9: **return** -1
 - 10: **else**
 - 11: **return** 1
 - 12: **end if**
-

Algorithm 3 demonstrates this prioritization. The Asimovian approach is service-oriented rather than democratic. Whereas utilitarian agents maximize aggregate welfare, Asimovian agents follow a rigid moral code with pre-defined rules. It does not harm non-Asimovian agents and defers to their commands but otherwise pursues its own survival, making it conditionally greedy. This partial restraint on greed outperforms unconditional greed while remaining cheaper to compute than full utilitarian deliberation.

VI. EVALUATION

We evaluate the effectiveness of ethical behavior as a heuristic for globally optimal behavior across a number of runtime performance and societal metrics. We show, based on runtime performance, that an omniscient leader agent coordinating globally optimal societal outcomes is computationally infeasible. We then compare outcomes for three kinds of Sugarscape artificial societies: one with greedy straightforward maximizers and two with ethically-aligned agents.

The Sugarscape environment was configured as shown in Figure 1. Any arbitrary cell held up to 4 sugar and spice. Every

cell regenerated 1 sugar and 1 spice per timestep up to its pre-configured maximum. Every agent has 12 potential movement choices each timestep. Behaviors like befriending, cultural transmission, disease transmission, lending, reproduction, and trading are dependent upon an agent’s movement choice.

We generated 500 random seeds used to create unique initial conditions. Each simulation ran for 2,000 timesteps and started with 250 agents scattered across the environment as shown in Figure 1, 50 of which were afflicted with unique diseases. Agents had a maximum lifespan between 60 to 100 timesteps, ensuring at least 20 generations of agents per simulation. The following agent features were enabled: combat, cultural transmission, disease transmission, friendship networks, lending, trading, and reproduction. Full details of these features can be found in previous work [17]. These agent features ensure agent lives are complex enough to make stable societies non-trivial to achieve while also remaining simple enough to aid in computational feasibility. Simply put, we demonstrate that acting on complete information in even a simplified decision space explodes in computational complexity.

VII. RESULTS

We initially consider the runtime performance of the omniscient leader and the 3 behavioral models under study. We demonstrate that the concept of a leader, or any other decisionmaking process across complete information, is computationally infeasible. We then consider the relative success of the 3 behavioral models to determine whether ethical behaviors perform appreciably better than straightforward maximization as a well-established, default behavior.

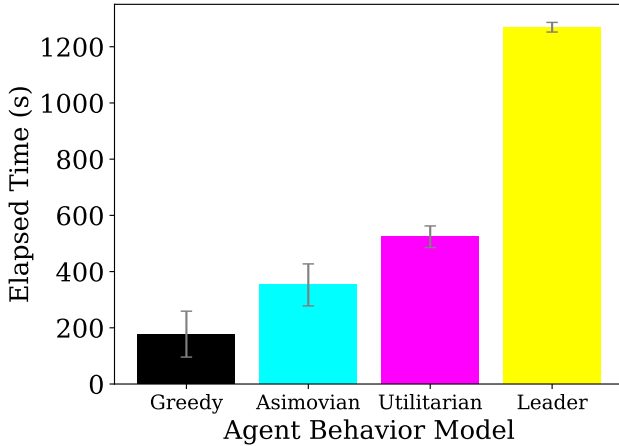


Fig. 3. Runtime for Agent Behavioral Models

A. Runtime

If a decisionmaking strategy takes too long, it will not matter if it is effective. The usefulness of a strategy is proportional to its timeliness to act upon the information it has. Figure 3 shows the mean runtime in seconds, with the standard deviation as error bars, for the leader agent approach and the 3 behavioral models under study. For the 3 behavioral models, the elapsed

time represents a complete execution of a Sugarscape run for 2,000 timesteps with 250 starting agents. For the leader approach, the elapsed time represents the consideration of 10,000 states in the first timestep (recall that the search space for timestep 1 is 12^{250} states).

The lengthy runtime for the leader approach is partially due to the implementation of the search. Since the decision space is vast yet each state is independent, it may be appealing to parallelize this search in order to achieve a certain runtime [19]. This approach requires enough memory to replicate complete information across all processes searching the decision space. Given the sheer number of states, this is not feasible for realistic memory sizes.

Instead, we minimize memory usage by considering only one state at a time. The leader agent in Sugarscape has an overhead of playing out a given state 1 timestep into the future, calculating the reward, and reverting the simulation back to the previous timestep in order to investigate another state. This slows down runtime to an infeasible degree. A parallelized or distributed approach would run faster, but there will be overheads for copying complete information to all searchers and for inter-process communication (to either synchronize results to the leader or to come to a consensus on the best move in a decentralized architecture).

In either case, the cost is extreme. Investigating one state at a time leads to a runtime measured in eons. Investigating most or all of the states in parallel has a memory demand measured in planetary-scale computing resources. Taking a middleground strategy will partially incur the downsides of both approaches, so the price of computational infeasibility is still paid.

TABLE I
SOCIETY SURVIVAL ACROSS 200 SEEDS

Agent Behavior	Extinct	Stable
Greedy (Default)	215 (43.0%)	285 (57.0%)
Asimovian	129 (25.8%)	371 (74.2%)
Utilitarian	5 (1.0%)	495 (99.0%)

B. Societal Survival

Having made the case for the infeasibility of a coordinating leader agent with complete information, we consider the relative success of the 3 behavioral models under study: greedy maximization, Asimov’s Three Laws of Robotics, and utilitarianism. The most crucial metric for success in a multi-agent system is its survival rate. If agent interactions lead to systemwide collapse, there is no way it can fulfill its mission. In Sugarscape, we assume for sake of argument that the leader agent’s coordination is informed by an objective function yielding optimal outcomes like perfect societal survival. Since it is infeasible to wait for the leader to find those optimal outcomes, we compare the outcomes of greedy maximizers and two models of ethical decisionmaking as heuristics for the highest possible performance.

Table I shows the survival rates for the three behavioral models under study across 500 seeds. Sugarscape societies either go extinct or become stable. Extinct societies had no

agents remaining by the end of 2,000 timesteps whereas stable ones ended with more than the 250 starting agents. No society was smaller at the end from where they started.

When agents behave greedily, maximizing their own interests, society becomes generally unstable. While Sugarscape societies make it more often than not, it is close to being a coin toss. This is unlikely to be acceptable odds for most multi-agent systems if productivity is desired.

When agents abide by ethical principles, societal survival greatly improves. Following Asimov's Three Laws of Robotics results in 17.2% higher stability due to principles of harm reduction and self-preservation, not self-maximization. Following Bentham's utilitarianism results in 99% stability for the experimental conditions due to explicit cooperation between agents. Neighborhoods of agents democratically determine movements, which takes slightly longer than the individualist Asimovian approach but yields far higher societal survival.

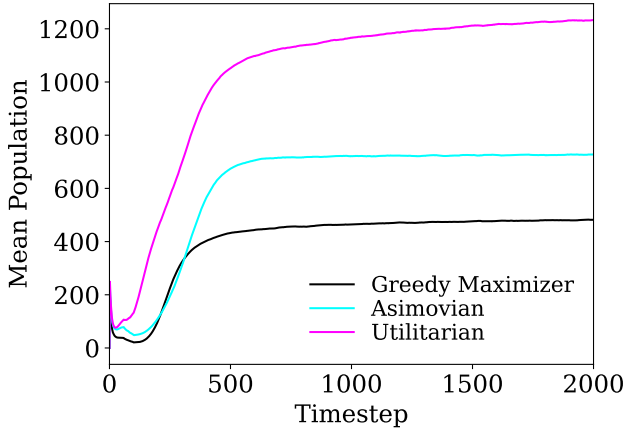


Fig. 4. Mean Population

C. Population

While survival is crucial for a multi-agent system to continue its mission, it is also important to consider how many agents stick around in these systems. In Sugarscape, agents can reproduce leading to increased system size over time which is not universal for multi-agent systems (where size may decrease monotonically). Figure 4 shows the mean population size per timestep across 500 seeds for the 3 behavioral models under study. For societies that go extinct, a zero is propagated for each timestep of a seed after the society goes extinct to avoid cherrypicking only the best results for each model. This affects the magnitude but not ordering of our results.

When greedy maximizers form a society, their population is diminished. Agents seeking straightforward maximization have no compunctions against killing other agents or taking resources that a starving neighbor agent needs. As a result, these societies are roughly a third the size of utilitarian ones. The Asimovian model yields roughly 75% larger population than the greedy one while the utilitarian model yields 200% higher. Though these two models embody radically different principles, they both yield large societies.

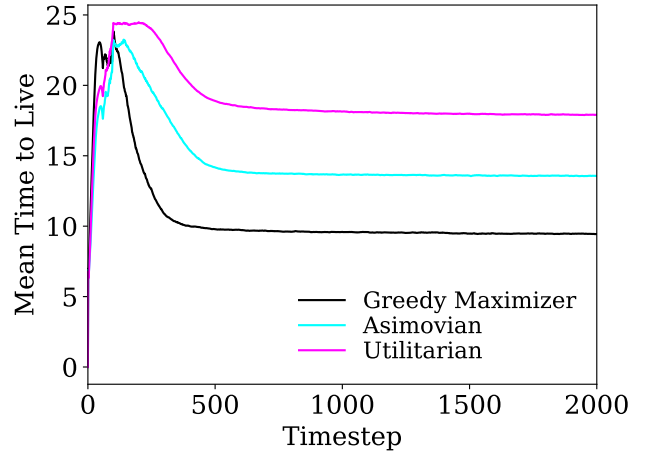


Fig. 5. Mean Agent Time to Live

D. Time to Live

Population does not indicate the economic success of any arbitrary agent. Time to live (TTL) is the number of timesteps an agent can survive given its metabolic needs and its current resource stockpile. It is an indicator of an agent's ability to survive periods of resource shortfalls.

Figure 5 shows the mean TTL per timestep across 500 seeds for the 3 behavioral models under study. Once the system stabilizes around 500 timesteps, the greedy maximizer strategy yields relatively low mean TTL. The Asimovian approach yields roughly 50% higher mean TTL, meaning Asimovian agents have enough resources to last 5 more timesteps without gathering more. Utilitarian societies have a mean TTL twice that of the greedy strategy. Mean TTL demonstrates the benefits of ethical behavior of short-sighted greed.

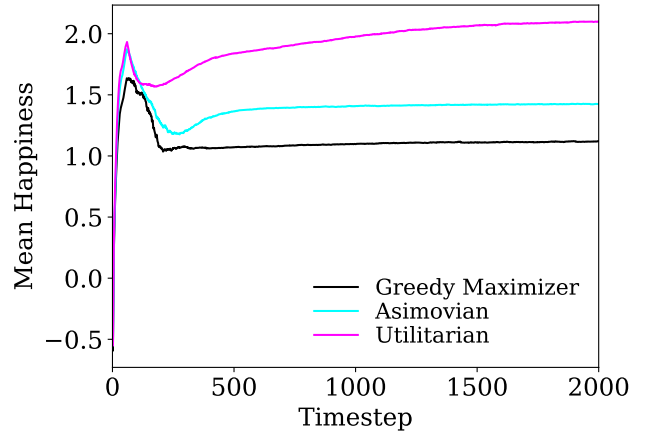


Fig. 6. Mean Agent Happiness

E. Happiness

Mean TTL provides context for the economic success of agents but conveys little about the overall prosperity of them. Agent happiness is a composite metric (ranging from -5.0 to

5.0) which consists of agent prosperity along five aspects: combat, family, health, social, and wealth. These aspects holistically represent an agent’s prosperity. An ideal score of 5.0 would result from having not killed another agent that timestep, having a large and living family, having no disease, having many friends, and holding a resource stockpile far above the average in society. The worst score of -5.0 results from the opposite of these. Specifics for how each happiness aspect is calculated can be found in previous work [20].

Figure 6 shows the mean happiness per timestep across 500 seeds for the 3 behavioral models under study. The two ethical models outperform greedy maximization. The Asimovian approach yields roughly 40% higher happiness while the utilitarian model yields about 110% higher happiness. Agents in these societies are significantly happier across multiple facets of the composite score than societies governed by greedy, straightforward maximization.

VIII. DISCUSSION

Our results demonstrate that moral deliberation is not merely a philosophical nicety but a practical strategy. Ethical agents consistently outperform greedy ones across each societal metric measured, and they do so without any coordinating authority. Each ethical agent reasons locally, from incomplete information. The artificial society that emerges from those individual deliberations outperforms one where the agents prioritize greedily taking whatever they can.

This points to a broader design principle: hierarchies of coordination over large state spaces are expensive, brittle, and fundamentally incapable of holding complete information about a dynamic environment. An agent that can ethically reason does not need to be told the right move to make. It approximates the globally optimal choice from its local vantage point, without communication overhead, staleness difficulties, or memory demands that plague any centralized architecture. Embedding moral reasoning directly into an agent makes large-scale autonomous systems more robust.

A. Limitations

Our conclusions are bounded by the ethical models we consider. We implement a utilitarian decision procedure as a representative consequentialist approach and Asimov’s Three Laws of Robotics as a rules-based approach. However, other ethical perspectives like social contract theory or virtue ethics may yield different outcomes we do not evaluate here.

Sugarscape agents have a mission of resource extraction and consumption. This limits how fully any ethical theory can be expressed. Moral considerations that have no analogue in the simulation, such as honesty or dignity, cannot meaningfully influence agent behavior.

Our notion of optimality reflects our chosen definition of societal success. A different reward function would relocate the global optimum that the leader pursues. We treat Sugarscape as a digital terrarium for machine ethics instead of a faithful model of human society as originally intended [16]. A different definition of success may render certain ethical perspectives incompatible with success and optimality.

B. Broader Applicability

While we use Sugarscape as a testbed for moral decisionmaking in multi-agent systems, our observations have applicability beyond it. Most importantly, the conceptual and performance limitations preventing effective use of complete information in Sugarscape hold true across multi-agent AI systems. Finding the optimal choice is essentially a graph search across all possible states. Sugarscape provides a relatively low complexity where each agent has up to 12 choices each timestep. Other multi-agent systems, like a robotic manufacturing floor or a smart highway, involve many agents with diverse, real-time choices.

The number of states exponentially increases with the number of possible actions per agent and the number of agents in the system. An exhaustive search guarantees finding the optimal choice while traditional heuristic approaches may prune the search space without guaranteeing optimality. The runtime can be marginally reduced by distributing the search space at the cost of memory usage. Even so, the sheer number of states to search for any realistic problem size is computationally intractable.

Ethical deliberation offers an alternative heuristic approaching optimality without requiring an exhaustive search over complete information. It is demonstrably superior to straightforward maximization which is myopic with regard to systemwide and individual agent outcomes. There are numerous AI systems for which moral decisionmaking is a significant benefit. Human-facing systems, like agentic AI which may require multiple agents to complete tasks, require alignment to human values to provide reasonable outputs. Autonomous medical systems, like robotic surgery or elderly care technologies, must adhere to moral guidelines for the medical profession, and algorithmified morality ensures that decisions are both timely and well-aligned to human expectations.

However, there are some systems for which our outlined ethical approach is not desirable. Our observations apply less directly to systems where a narrow definition of optimality is both achievable and required, such as safety-critical control systems with fully-observable state. Systems where self-interest is the correct objective by design would also not be as applicable here, for example a competitive environment of game-playing agents or autonomous stock trading.

IX. REPRODUCIBILITY OF EXPERIMENTATION

Sugarscape is deterministic, meaning our presented results are reproducible for the v2026.3 release.¹ We provide the required configuration files to reproduce the dataset² in the `follow-the-leader.zip` archive. To run the simulation for a single configuration file, execute the following (substituting the system’s Python 3 alias):

```
> python sugarscape.py -c <CONFIG>
```

¹Software at: <https://github.com/digital-terraria-lab/sugarscape>.

²Configuration files at: <https://github.com/digital-terraria-lab/datasets>.

The `numParallelSimJobs` property located in `config.json` specifies the number of simulations allowed to run simultaneously. Unzip and place the dataset configuration files into the data directory before reproducing the results. Run `make setup` to check for the local Python 3 installation. Run `make data` to start data collection. Run `make plots` to generate figures. Minor tweaks to `plot.py` in the `plots` directory are needed to reproduce exact figure formatting. To reproduce the dataset and produce figures, run:

```
> make setup
> make data
> make plots
```

X. CONCLUSIONS AND FUTURE WORK

We showed that globally optimal decisionmaking in multi-agent systems is computationally infeasible even at the modest scale of Sugarscape. Our omniscient leader coordinating 250 agents faced a search space of 12^{250} states which made exhaustive search intractable regardless of hardware. In place of this unreachable optimum, we demonstrated that two ethical decisionmaking frameworks serve as tractable heuristics for optimal behavior. Both require only local information, run in an infinitesimal fraction of the time the leader spends, and produce substantially better systemwide outcomes than straightforward maximization. The Asimovian approach led to roughly 17% higher system survivability, 75% larger populations, 50% higher economic success, and 40% higher agent prosperity. The utilitarian approach yielded about 42% higher system survivability, 200% larger populations, 100% higher economic success and 110% higher agent prosperity.

A natural direction for future work is to broaden the repository of ethical decisionmaking algorithms that researchers and practitioners can deploy into multi-agent systems as they choose. The key challenge lies in faithful translation of ethical reasoning into concrete parameters that can be computed by an agent at runtime. Each translation is a contribution in its own right, and a growing repository of such translations would make transparent, explainable, and ethical agent design accessible across a wide range of multi-agent systems.

Another fruitful avenue of investigation is the architectural gap between a single coordinating leader and fully-independent agent decisionmaking. Intermediate architectures may capture benefits of both coordination and autonomy without the infeasibility of complete information. An example would be a mechanism for agents to elect local neighborhood leaders rather than a single, global one. We leave systematic exploration of these hybrid architectures for future work.

XI. ACKNOWLEDGEMENTS

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